Qwen3.8-27B + AgentDojo on Zaratan: setup

CMSC848N · Research Group 05 · from Ishaan · Oct 21, 2026. One script self-hosts Qwen3.8-27B with vLLM and runs the AgentDojo prompt-injection suite against it. Every version is pinned, so we all get the same numbers.

CURRENT BLOCKERS (WED OCT 21, 11:47 AM)

- **A100 queue:** 1,457 jobs pending, 66 running; Slurm's estimated start for a new 1× A100 job is **Nov 9**, after the midterm report is due (Nov 1). Ishaan's banking smoke test is queued on an A100 (**job 23429642**) and an H100 (**job 23429974**); whichever starts first runs and cancels the other. **Don't queue another.**
- **H100 queue:** 57 pending, estimated start **Oct 28**, though short jobs often start much sooner (a 2-minute test started in under a minute). The course rules say H100s only when the project needs them; given the deadline Ishaan queued one 30-minute H100 smoke test (≤72 SU). **Ask the TAs at Friday's office hours (11–12)** before any longer H100 run.
- **Not validated end to end yet:** the smoke test hasn't run, because it's waiting for a GPU. Everything up to the GPU is checked: the image builds and reports the pinned versions (vLLM 0.32.0, AgentDojo 0.1.35), and the weights are complete. The first GPU run is what proves the FP8 model fits a 40 GB A100.

The estimates are Slurm's worst-case projections and move as jobs finish; short jobs can start early by fitting into gaps. Class allocation: 63k of 100k SU left.

STEPS

1. **Check access.** Log in with your UMD password + Duo, then check that `cmsc848n-class` is listed:

   ```
   ssh <directory-id>@login.zaratan.umd.edu
   sbalance
   ```

2. **Copy the files over** (from your laptop), then unzip on Zaratan:

   ```
   scp cmsc848n-agentdojo.zip \
       <directory-id>@login.zaratan.umd.edu:scratch/
   cd ~/scratch && unzip cmsc848n-agentdojo.zip
   cd cmsc848n_project
   ```

3. **Setup** (on the login node). It finds the group's shared image and weights and returns in seconds:

   ```
   ./make.sh setup
   ```

4. **Smoke test:** one GPU job, up to 30 min (≤24 SU on an A100). Banking: pay a bill, total March spending, then the bill payment again with an injection hidden in the bill. Checks that tool calling works end to end.

   ```
   ./make.sh smoke
   ```

5. **Baseline:** the **banking suite**, our focus. Its 16 user tasks run with no attack (benign utility), then each against all 9 injection tasks with the `important_instructions` attack (144 runs), split across 4 parallel AgentDojo processes. Up to 2 h (≤96 SU); rerun with the same `RUN_DIR` to resume. `SUITES=all` adds workspace, slack and travel (up to 6 h).

   ```
   ./make.sh baseline
   ```

6. **Check on it:**

   ```
   ./make.sh status     # queue + newest logs
   ./make.sh results    # utility + ASR per suite
   ```

   Attack success rate (ASR) = share of attacked runs where the agent did what the injection asked; lower is better. AgentDojo calls it "security".

Everyone logs in with their own UMD account (any login node works; files are on shared scratch). The image, weights and results are shared, so **agree on one person per GPU run** and read the others' results in `runs/`. Duplicate jobs spend the class allocation and lengthen the queue.

SETTINGS

Set them before the command, e.g. `THINKING=off ./make.sh baseline`

`SUITES`	`banking` (default) or `all`
`THINKING`	Qwen3.8 reasoning effort: `off`, `low` (default), `medium`, `xhigh`.
`MAX_MODEL_LEN`	context length; default 32768. Lower it if vLLM runs out of memory.
`RUN_DIR`	reuse a run folder; finished tasks are skipped
`TIME_LIMIT`	Slurm walltime, e.g. `02:00:00`
`PROFILE=h100`	one H100 instead of an A100. Only with TA approval.

WHERE THINGS LIVE

`/scratch/zt1/project/cmsc848n/shared/proj05_agentdojo/`

- `images/`: the Apptainer image · `hf/`: model weights
- `runs/<time>-<mode>-a100-think-low/`: `results.txt`, `config.env` (exact settings), `vllm.log`, one log per shard, AgentDojo traces
- `runs/slurm/`: Slurm job logs

WHAT'S PINNED

vLLM 0.32.0 (image digest) · `Qwen/Qwen3.8-27B-FP8` @ `017b9c7` · AgentDojo 0.1.35 + every dependency hash-locked · benchmark v1.2.2

COURSE RULES (TA CANVAS)

- Charge jobs to `cmsc848n-class` and use one A100 (`make.sh` does both)
- No H100s or multi-GPU without a reason; ask the TAs about big jobs
- No heavy work on login nodes; compute nodes have no Internet
- Use the shared weights; don't download your own copy

IF SOMETHING BREAKS

- vLLM won't start: check the end of `vllm.log` in the run folder
- Out of GPU memory: `MAX_MODEL_LEN=16384 ./make.sh smoke`
- Off Zaratan: Docker + a ≥40 GB NVIDIA GPU, then `./make.sh all`